HF Channel Sounder — Over-the-Air Test Results

Hermes-Lite 2 SDR • ~90-mile ionospheric link • 7.13 MHz • 2026-06-21

Bottom line

Method (OTA)	Result	Matched-filter performance
LFM chirp	SUCCESS	**+26.8 dB** processing gain — raw –7.0 dB → **+19.8 dB** SNR
DSSS / PN	SUCCESS	**+24.1 dB** processing gain — raw –5.2 dB → **+18.9 dB** SNR
OFDM	FAILED	did not decode over the air (carrier-offset sensitive)

Two independent methods agree, so the result is trustworthy: a spread-spectrum / coded waveform **closes a link that a bare tone cannot**. The matched filter lifts the raw sub-noise signal (–5 to –7 dB) to a usable **+19 dB**.

Channel characterization (over the air)

RMS delay spread	Coherence BW	Doppler spread	Coherence time
~1.9 ms	**~85 Hz**	**~3 Hz**	**~60 ms**

Delay spread = multipath (multiple ionospheric layers, O/X split, one- vs multi-hop). Doppler = the ionosphere's motion. Measured fading during the capture was mild (±1.8 dB), but a single 5 s snapshot — expect deeper fades at other times.

Frequency offset — why OFDM failed

The two radios run on **independent clocks**. The CW tone measured a steady –**3.15 Hz carrier-frequency offset** between them (plus a slow sample-clock drift). OFDM is acutely sensitive to this: with no offset correction, the subcarrier phases rotate and the symbol timing drifts, so the known pilots no longer line up and the constellation collapses — **EVM 607%, did not decode**.

The matched-filter methods (chirp, DSSS) are inherently tolerant of a frequency offset — it merely shifts the correlation peak — which is why they succeeded on the same link at the same moment.

Recommended radio design

Build the data link on a **DSSS-style spread-spectrum waveform with strong FEC**. The test shows it carries text where a plain tone (–5 dB) cannot. Anticipated steps to handle the three real impairments this path showed:

- **Fading (multipath, ~60 ms coherence time):** interleave the FEC-coded bits over more than ~60 ms so a fade corrupts scattered bits the code can repair; the spreading gain holds shallow fades above noise; add ARQ (retransmit) for the deepest dropouts.

- **Frequency drift / offset (~3 Hz between radios):** add a preamble for coarse carrier-frequency acquisition and a pilot or tracking loop for fine correction; favor offset-tolerant modulation (DSSS / chirp / MFSK). The ultimate fix is GPS-disciplining both oscillators to a common reference.
- **Phase noise (oscillator instability):** use differential or non-coherent demodulation (DBPSK / DQPSK, or MFSK like FT8) so detection does not depend on absolute phase; alternatively pilot-aided phase tracking; keep symbols short relative to the phase-noise timescale.

Generated from HL2 HF Sounder captures (xSounder_Tool), RF 7.13 MHz, fs 48 ksps, LNA 0 dB, 5 s grabs. Methods cross-checked: CW, LFM chirp, DSSS/PN, OFDM (loopback and over-the-air).